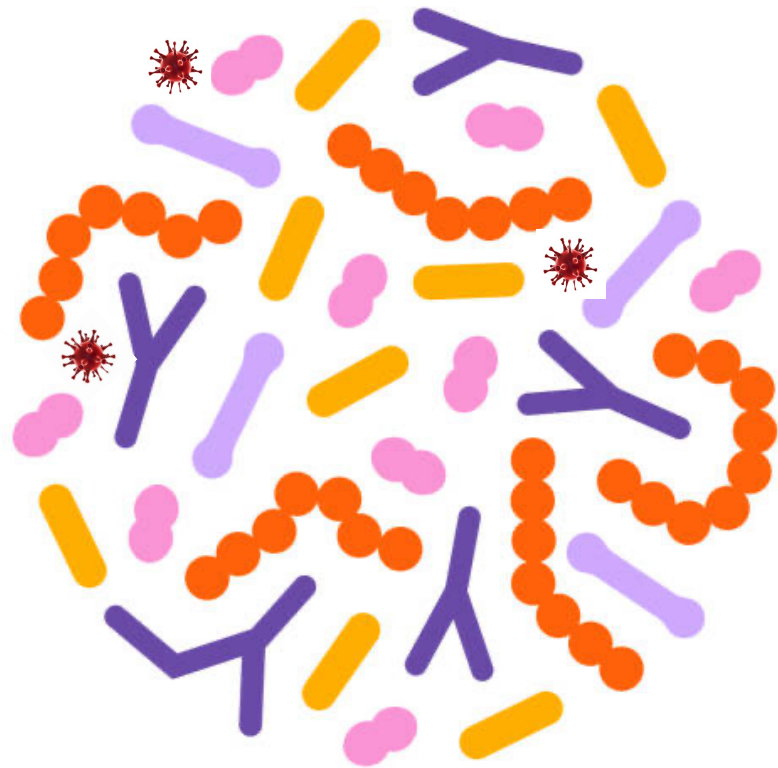




Processing of shotgun metagenomics samples

- nf-core/taxprofiler together with Metaphlan v4 database
- Genome-level marker gene based taxonomic classifier
- Data Science platform
- BRIGHT
- Spring Data-type Courses
- 6th May 2026

Metagenomic insights into the prokaryotic communities of heavy metal-contaminated hypersaline soils

Cristina Galisteo^a, Fernando Puente-Sánchez^b, Rafael R. de la Haba^a, Stefan Bertilsson^b, Cristina Sánchez-Porro^a,
Antonio Ventosa^a  

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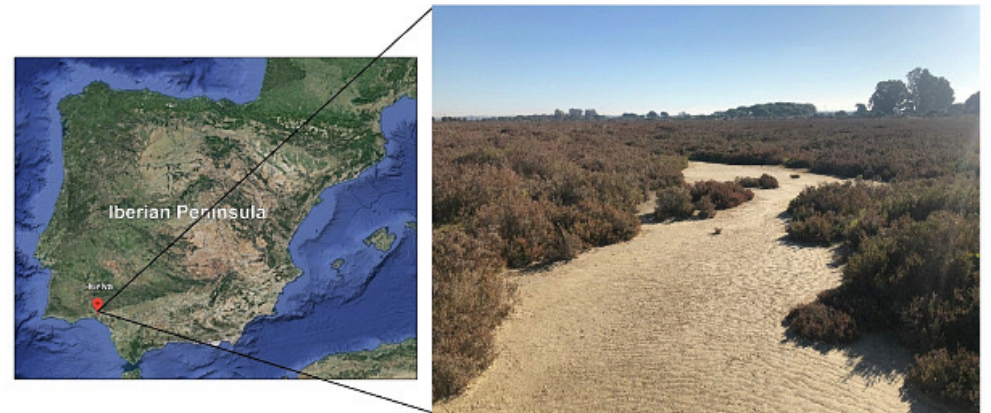
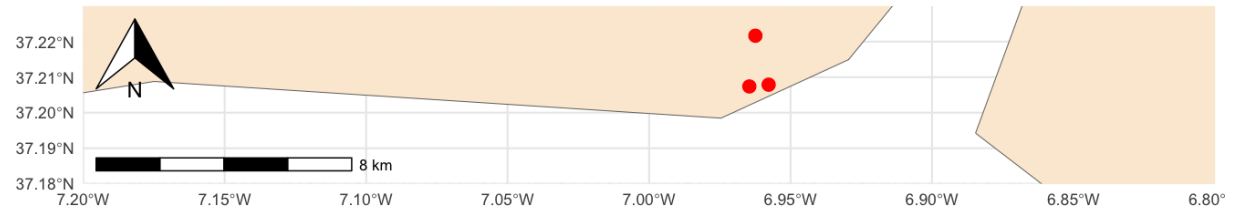
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- Saline soils and their microbial communities have recently been studied in response to ongoing desertification of agricultural soils caused by anthropogenic impacts and climate change.
- In this study, they describe the prokaryotic microbiota of hypersaline soils in the Odiel Saltmarshes Natural Area of Southwest Spain.
- This region has been strongly affected by mining and industrial activity and feature high levels of certain heavy metals.
- **18** shotgun metagenomes sequenced with Illumina NovaSeq from three different areas in 2020 and 2021.

Sampling sites

- Three sampling sites (1, 2 and 3) were selected in order to obtain a representative subset of the prokaryotic community of the soils from the Odiel Saltmarshes Natural Area



Sample table with some metadata:

- **Sampling dates (2)**
- **Areas (3)**
- Coordinates
- Sample names (n = 18)

Sampling	Date	Area	Samples	Coordinates
M2	July 2020	1	M2_1A	37°12'26.6"N 6°57'52.5"W
			M2_1B	
			M2_1C	
		2	M2_2A	37°12'28.4"N 6°57'27.9"W
			M2_2B	
			M2_2C	
		3	M2_3A	37°13'18.0"N 6°57'44.8"W
			M2_3B	
			M2_3C	
M3	June 2021	1	M3_1A	37°12'26.6"N 6°57'52.5"W
			M3_1B	
			M3_1C	
		2	M3_2A	37°12'28.4"N 6°57'27.9"W
			M3_2B	
			M3_2C	
		3	M3_3A	37°13'18.0"N 6°57'44.8"W
			M3_3B	
			M3_3C	

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A00180 → instrument name

10 → run id

H7NK5DMXX → flowcell id

1 → flowcell lane

1101 → tile number within the flow cell line

16821 → x-coordinate of the cluster within the tile

1000 → y-coordinate of the cluster within the tile

SPACE

1 → member of the pair (if paired-end)

N → N if read is not filtered, Y if it is filtered

0 → when none of the control bit are on

GGACTT, GTCGTTTCG → barcodes are used for demultiplexing and assign reads to the correct sample



Four lines for each read:
 1. identifier (starting with @)

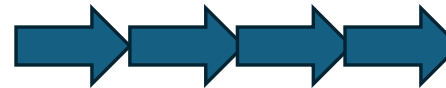
1. identifier (starting with @)
2. Read sequence
3. a line starting with + (sometimes id gets repeated)
4. Phred quality scores representing the base-caller's confidence – equal in length to the read sequence

[illegible]

Repeating steps too many times? – find yourself a pipeline!

```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTTCG
TGCAGCAGCTAATGAGGAACCACTTCTCCCTCCAGCGCTCTAAATACCTCAGAACAATAGGATCATCATAAATCCCCTAGTCTGAAC TG
+
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTTCG
TGCAGAAGAAGAAAGCACAAAGTATTACGCCTATCCTTCATATTTCCGCAAGGTAACATATCTCGGTTTCATATCGAGATTATATAGAATCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTTCG
TGCAGGAAGTTATGCAGGGGATCCTGTATTATTAATAGAGCACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG
+
FF-88F8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTTCG
TGCAGAGTACATCAACAAAAGAAACCTAACTGCCCTACCGGCAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCAGTCAATATAAGG
+
8F888FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
```

Series of steps in
an specific order



Abundance table for
downstream analysis

OTU	Sample 1	Sample 2	Sample 3
A	0.60	0	0.70
B	0.24	0.20	0.10
C	0.10	0	0
D	0.05	0	0
E	0.01	0	0
F	0	0.80	0.20
Total	1.0	1.0	1.0

- Different steps to run manually in an order (QC, host removal, mapping to a database, etc...)
- Workflow that contains all the processes needed and have them chained



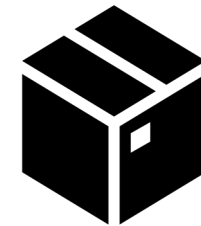
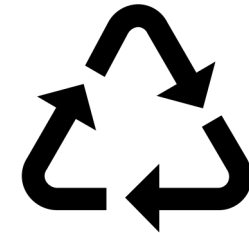
 nextflow

Nextflow is a language to build workflows and a software to orchestrate workflows execution.

- Strong community behind with various bioinformaticians joining forces to build each of the pipelines
- Support for other technologies

Nextflow's core features

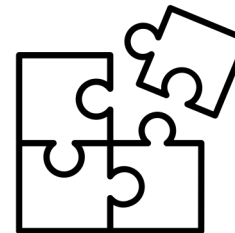
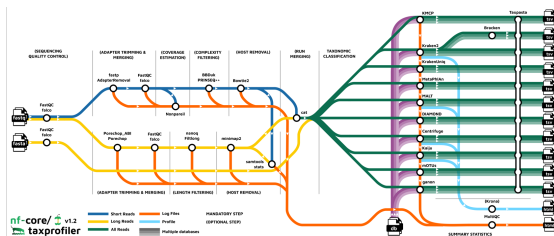
- **Reproducible** between runs
 - integration with code management tools
 - all packages downloaded, organized in containers, and control over computing environment
- **Portable** between systems
 - you can write the code in your laptop and can run everywhere (HPC, cloud)
 - works with most of computing environments
- **Scalable**
 - it can run for 10 samples on your laptop or thousands in the cloud
- **Integration** of existing tools, systems, and industry standards





A global community effort to collect a curated set of open-source analysis pipelines built using Nextflow.
<https://nf-co.re/>

<https://www.nature.com/articles/s41587-020-0439-x>



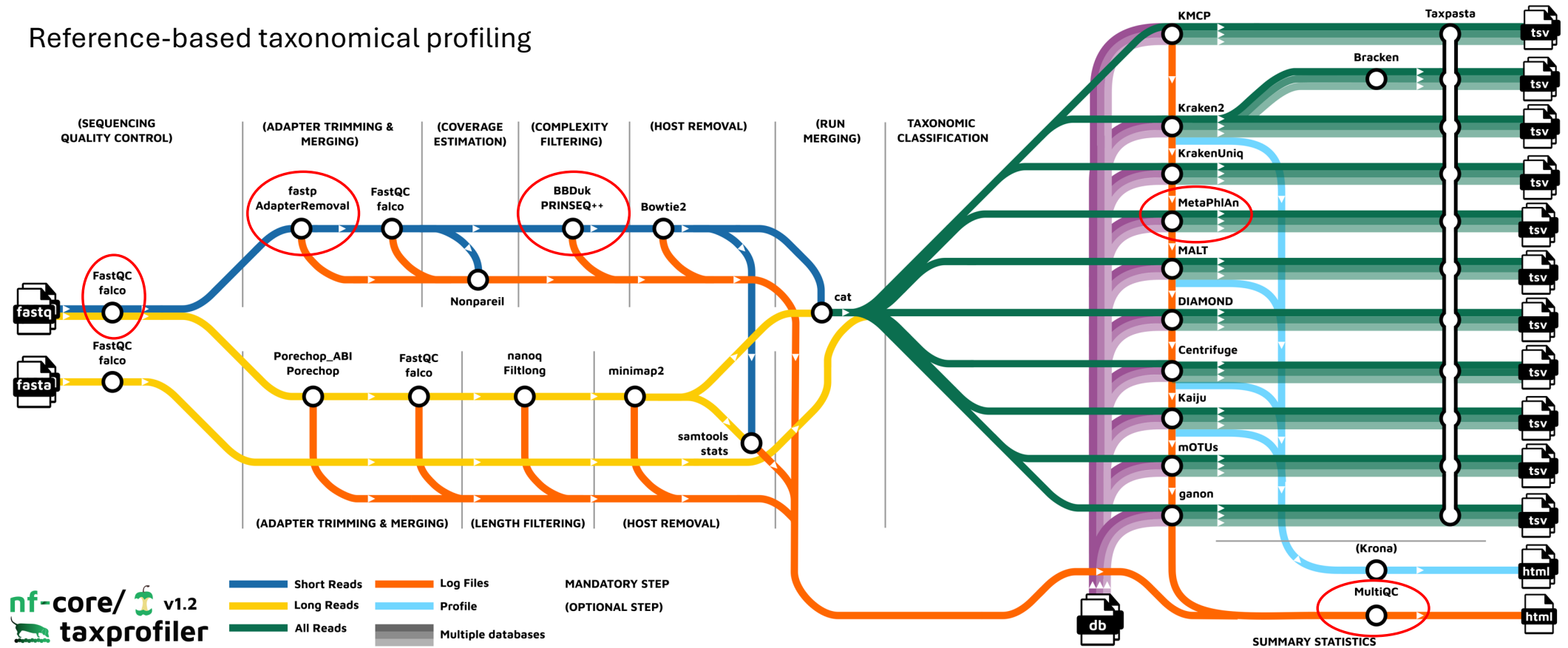
149 pipelines available
(2nd May 2026)
for 149 different
bioinformatic analysis

Modules: 1,811
reusable
components that
can be integrated
into pipelines

Subworkflows: 117 pre-
assembled
combinations of
modules aimed at
streamlining commonly
used workflows

nf-core/taxprofiler

Reference-based taxonomical profiling



Command line to run the pipeline

➤ nextflow run nf-core/taxprofiler -profile test,docker --outdir processed_data

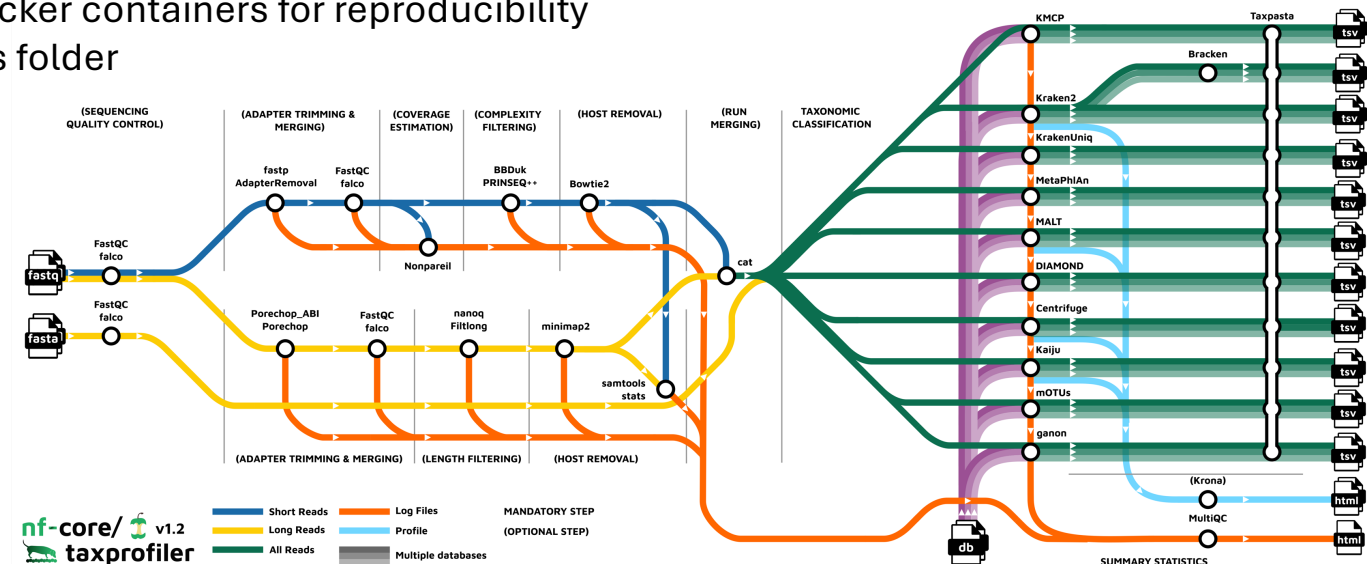
nf-core/taxprofiler → nf-core pipeline downloaded from here:

<https://github.com/nf-core/taxprofiler/tree/2.0.0>

“test” → running for demonstration few reduced samples, each pipeline has a test dataset (in a github repo, so we do not have to specify the input files this time)

“docker” → using Docker containers for reproducibility

--outdir → our results folder



Wanna know more about Nextflow: https://biosustain.github.io/dsp_nextflow_training/